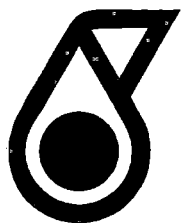


**SPECIAL CORE ANALYSIS OF ROTARY
SIDEWALL CORES**

DEPTH: 3184.0, 3528.0, 3576.0, 3598.5 M

**DUYONG DEEP-1
MALAYSIA**

PETRONAS CARIGALI SDN. BHD.



PETRONAS

SPECIAL CORE ANALYSIS

of

ROTARY SIDEWALL CORES

from

WELL DUYONG DEEP-1

Depths: 3184.0, 3528.0, 3576.0, 3598.5 m.

for

PETRONAS CARIGALI Sdn. Bhd.

Block PM12

Offshore Peninsular Malaysia

by

UZMA ENGINEERING Sdn. Bhd.

Report No.: LB-00-32-SCAL



PT. GEOSERVICES

Geological & Laboratory Services Division

in association with

PT. GEOSERVICES

Geological & Laboratory Services Division

Report No.: 2014/0826/LAB



Restor



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I. INTRODUCTION

a. Summary

This report provides the results of special core analysis for rotary [MSCT] sidewall cores from **Well Duyong Deep-1**, from the interval 3184.0-3598.5 m.

The well was drilled by **PETRONAS CARIGALI Sdn. Bhd [PCSB]** in **Block PM12**, offshore Peninsular Malaysia.

The analyses were undertaken and report prepared by **UZMA ENGINEERING Sdn. Bhd. [Uzma]** in partnership with **PT. GEOSERVICES, Geological & Laboratory Services Division [PTG]**.

b. Objectives

The object of the study was to carry out the following analyses:

- Formation resistivity factor at net overburden [NOB] conditions
- Formation resistivity index by continuous injection at NOB conditions
- Cation exchange capacity
- Air-brine capillary pressure by centrifuge at NOB conditions
- Mercury injection capillary pressure to maximum of 2,000 psi

c. Samples & Analyses

Six [6] rotary sidewall cores were received for routine core analysis and rock description [petrography, XRD and SEM] and the results are presented in separate reports. Four [4] of the samples were then selected, by **PCSB**, for special core analysis, as listed in *Figures 1-2*.

Analyses	Method	Sample meters	No.
SPECIAL CORE ANALYSIS	Formation factor at NOB conditions	3184.0 3528.0 3576.0 3598.5	4
	Resistivity index by injection method at NOB conditions		
	Air-brine drainage capillary pressure by centrifuge at NOB conditions		
	Cation exchange capacity		
	Mercury injection capillary pressure to max. of 2,000psi		

**FIGURE 1: WELL DUYONG DEEP-1
SUMMARY OF ANALYSES**



Sample ID		Formation factor at NOB conditions [4]	Resistivity index through continuous injection at NOB conditions [4]	Cation exchange capacity [4]	Air-brine drainage capillary pressure by centrifuge at NOB conditions [4]	Mercury injection capillary pressure to max of 2,000 psi [4]
Sample No.	Depth meters					
2	3184.0	√	√	√	√	√
3	3528.0	√	√	√	√	√
4	3576.0	√	√	√	√	√
6	3598.5	√	√	√	√	√

**FIGURE 2: WELL DUYONG DEEP-1
SAMPLES & ANALYSES**

d. Analysts

Special core analysis : Soenardi, Trimudi Hartono
Co-ordination : Bob Meyrick, Soenardi
Logistic & technical support by UZMA : Robert Hulsbos, Gerard Raj



II. SPECIAL CORE ANALYSIS

a. Introduction

Results are provided for the following on the four [4] samples selected for analysis:

- Formation factor at NOB conditions.
- Resistivity index by continuous injection at NOB conditions.
- Cation exchange capacity.
- Air-brine drainage capillary pressure by centrifuge at NOB conditions.
- Mercury injection drainage capillary pressure.

b. Results

- **Formation Resistivity Factor & Resistivity Index at Net Overburden Conditions:**

Tables 1, 2, Graphs 1, 2.

The individual cementation exponent “m” ranged from 1.83 to 1.85 with an average value of 1.84 for the four [4] samples.

The individual saturation exponent “n” ranged from 1.87 to 1.90 with an average value of 1.88 for the four [4] samples.

- **Cation Exchange Capacity:**

Table 4.

The cation exchange capacity ranged from 1.0800 to 1.1600, with an average of 1.1150 meq / 100 gr, for the four [4] samples analyzed.

- **Air-Brine Drainage Capillary Pressure by Centrifuge Method at NOB Conditions:**

Table 3, Graph 3.

An irreducible water saturation of 83.7% of pore volume was measured at the end of drainage testing for the sample at 3184 m.

Water saturation measurements could not be made on three [3] of the samples which were very tight.

- **Mercury Injection Capillary Pressure and Pore Size Distribution to Maximum of 2,000 psi :**

Figure 3, Tables 5a-d, Graphs 4a-d, 5a-d, 6a-d.

The pore size distributions for the samples were calculated by dividing the range of pore throat sizes [r] into three [3] primary groups, with an average of macro 15.8%, meso 1.0% and micro 83.2%.



Sample ID	Depth meters	Porosity %	Pore Size Distribution		
			Micro R<0.5	Meso 0.5<r<1.5	Macro r>1.5
2	3184.0	11.3	88.9	1.4	9.7
3	3528.0	9.3	86.2	0.8	13.0
4	3576.0	11.2	77.6	1.7	20.7
6	3598.5	10.0	80.4	0.0	19.6

**FIGURE 3: WELL DUYONG DEEP-1
SUMMARY OF PORE SIZE DISTRIBUTION**

c. Laboratory Procedures

The following are generalized procedures involved in the measurements provided.

Routine core analysis measurements used in this report are those described in earlier report by *UZMA/PTG*.

- **Samples**

Four [4] rotary sidewall core samples were selected for the SCAL measurements requested by *PCSB*.

- **Fluid Preparation**

Synthetic formation brine [3,500 ppm] was prepared, as requested by *PCSB*, using reagent grade chemicals and de-ionized water. The fluid was evacuated of air and pre filtered through a 0.45 micron millipore™ prior to use.

- **Sample Saturation**

The samples were initially loaded into a saturation cell and evacuated for a minimum of 24 hours. The cell was filled with simple formation brine 3,500 NaCl ppm. The pressure was increased to 2,000 psig and maintained for a minimum of two [2] hours. The saturated samples were removed from the cell and weighed, and the gravimetric saturated pore volume was calculated and compared to the gas expansion pore volume to verify complete saturation.

- **Formation Resistivity Factor at Net Overburden Conditions**

The saturated samples were placed in individual hydrostatic core holders with a net overburden pressure of 560 psi. The downstream and upstream face of each sample was in



contact with a silver membrane and silver end-stem. The formation resistivity factor F_R at each pressure was calculated from each sample from resistance measurements of the brine and the brine saturated plugs. The Archie porosity [cementation] exponent “m” was calculated from the slope of the composite formation resistivity factor-porosity relationship, with values based on a Y-intercept value “a” of 1.00.

$$F = \frac{a}{\phi^m}$$

Where:

F = Formation factor
a = Intercept
m = Porosity exponent
 ϕ = Porosity

- ***Resistivity Index by Continuous Injection Method at NOB Conditions***

The formation resistivity index, I_R , was measured by the continuous injection method, in which light oil is injected at very low rates. Effluent water was monitored and saturation was calculated. The Archie saturation exponent “n” was calculated from the resistivity index-saturation relationships.

$$RI = \frac{a}{S_w^n}$$

Where:

RI = Resistivity index
a = Intercept
 S_w = Water saturation
n = Saturation exponent

- ***Cation Exchange Capacity***

The end trim samples were cleaned using toluene and subsequently dried then crushed and ground to a size of less than the 60-mesh sieve. The powder was then measure cation exchange capacity on each sample by standard titration method.

The value of Q_v was calculated using the following equation:

$$Q_v = \frac{CEC (1-\phi) GD}{\phi * 100}$$

Where: Q_v = Quantity of clay exchangeable cation present
CEC = Cation exchange capacity as meq/100 grams
 ϕ = Porosity, fraction
GD = Grain density, gr/cc



- ***Air-Brine Capillary Pressure by Centrifuge at NOB Conditions***

The saturated samples were loaded to centrifuge buckets at ambient temperature. They were spun at increasing incremental speeds equivalent to pressures of 1.0, 2.0, 4.0, 8.0, 15.0, 25.0, 50.0, 100.0 and 200psi.

Effluent brine volumes were monitored at each incremental pressure until stable. Average brine saturations were determined volumetrically then adjusted to end-face saturations using the **Forbes** calculations.

- ***Mercury Injection Capillary***

One quarter [1/4] inch end trims were taken from four [4] plugs for mercury injection capillary pressure measurements.

The samples were placed in a **Micromeretics Autopore IV** porosimeter, under a 50 micron vacuum and injected with mercury at about 33 increasing capillary pressure steps until 2,000 psi mercury injection pressures was obtained.

Results are summarized in the summary of mercury injection capillary pressure and pore size distribution data are shown in *Figure 3*.



TABLES



Saturant 3,500 ppm
Resistivity of saturant 1.500 ohm-m @ 77° F
Net overburden pressure 560 psi

Sample ID	Depth meters	Horizontal		Formation Resistivity Factor F	Porosity Exponent m
		Permeability to Air at NOB md	Porosity at NOB Fraction		
2	3184.0	0.134	0.116	54.72	1.85
3	3528.0	0.011	0.091	81.05	1.83
4	3576.0	0.051	0.116	53.95	1.85
6	3598.5	0.037	0.101	68.07	1.84

**TABLE 1: WELL DUYONG DEEP-1
FORMATION RESISTIVITY FACTOR AT NOB CONDITIONS**



Saturant 3,500 ppm
Resistivity of saturant 1.500 ohm-m @ 77° F
Net overburden 560 psi

Sample ID	Depth meters	Horizontal		Formation Resistivity Factor F	Porosity Exponent m	Brine Saturation Fraction V_p	Formation Resistivity Index I	Saturation Exponent n
		Permeability to Air at NOB md	Porosity at NOB Fraction					
2	3184.0	0.134	0.116	54.72	1.85	1.000	-	-
						0.933	1.139	1.89
						0.907	1.205	1.90
						0.884	1.263	1.90
						0.867	1.310	1.89
3	3528.0	0.011	0.091	81.05	1.83	1.000	-	-
						0.939	1.126	1.88
						0.920	1.169	1.88
						0.908	1.197	1.87
						0.896	1.228	1.87
4	3576.0	0.051	0.116	53.95	1.85	1.000	-	-
						0.933	1.139	1.87
						0.910	1.194	1.88
						0.892	1.241	1.89
						0.874	1.289	1.89
6	3598.5	0.037	0.101	68.07	1.84	1.000	-	-
						0.939	1.126	1.88
						0.925	1.159	1.89
						0.912	1.190	1.88
						0.885	1.260	1.89

**TABLE 2: WELL DUYONG DEEP-1
FORMATION RESISTIVITY FACTOR & RESISTIVITY INDEX AT
NOB CONDITIONS**



Pressure, <i>psi</i>														
					0	1	2	4	8	15	25	50	100	200
Sample d	Depth <i>meters</i>	NOB <i>psi</i>	Horizontal		Brine Saturation, <i>percent pore space</i>									
			Permeability to Air <i>md</i>	Helium Porosity %										
2	3184.0	560	0.134	11.6	100.0	100.0	100.0	100.0	100.0	100.0	100.0	97.2	92.1	83.7
3	3528.0	560	0.011	9.1	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
4	3576.0	560	0.051	11.6	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
6	3598.5	560	0.037	10.1	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

**TABLE 3: WELL DUYONG DEEP-1
AIR-BRINE CAPILLARY PRESSURE BY CENTRIFUGE**



Sample ID	Depth <i>meters</i>	Porosity <i>%</i>	Grain Density <i>gr/cc</i>	CEC <i>Meq/100gr</i>	Qv <i>Meq/ml</i>
2	3184.0	11.6	2.66	1.1000	0.2230
3	3528.0	9.1	2.69	1.1600	0.3117
4	3576.0	11.6	2.67	1.0800	0.2198
6	3598.5	10.1	2.69	1.1200	0.2682

**TABLE 4: WELL DUYONG DEEP-1
CATION EXCHANGE CAPACITY**



Sample ID 2
Depth 3184.0 meter
Permeability to air 0.108 md
Porosity 0.113 fraction V_p

Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction V_p	J Function
	fraction V_p	fraction V_b		A/B [Lab]	O/B [Lab]	O/B [Res]		
0.5	0.0000	0.0000	211.2451	0.0988	0.0577	0.0357	1.0000	0.0003
1.0	0.0000	0.0000	109.9337	0.1899	0.1108	0.0686	1.0000	0.0006
1.5	0.0000	0.0000	72.7939	0.2868	0.1673	0.1036	1.0000	0.0008
2.1	0.0000	0.0000	51.3024	0.4070	0.2374	0.1470	1.0000	0.0012
3.1	0.0000	0.0000	34.9789	0.5969	0.3482	0.2156	1.0000	0.0018
4.1	0.0000	0.0000	26.5357	0.7869	0.4590	0.2841	1.0000	0.0023
5.6	0.0000	0.0000	19.3768	1.0776	0.6286	0.3891	1.0000	0.0032
7.0	0.0000	0.0000	15.3250	1.3625	0.7948	0.4920	1.0000	0.0040
8.5	0.0000	0.0000	12.6301	1.6532	0.9644	0.5970	1.0000	0.0049
10.5	0.0000	0.0000	10.2410	2.0389	1.1893	0.7363	1.0000	0.0060
13.0	0.0000	0.0000	8.3001	2.5156	1.4675	0.9084	1.0000	0.0074
16.0	0.0000	0.0000	6.7334	3.1009	1.8089	1.1198	1.0000	0.0091
20.0	0.0000	0.0000	5.3975	3.8684	2.2566	1.3969	1.0000	0.0114
23.0	0.0000	0.0000	4.6923	4.4499	2.5957	1.6069	1.0000	0.0131
25.0	0.0000	0.0000	4.3129	4.8413	2.8241	1.7483	1.0000	0.0143
29.9	0.0000	0.0000	3.6008	5.7988	3.3826	2.0940	1.0000	0.0171
43.0	0.0000	0.0000	2.5072	8.3280	4.8580	3.0073	1.0000	0.0245
43.0	0.0000	0.0000	2.5066	8.3299	4.8591	3.0080	1.0000	0.0245
56.4	0.0000	0.0000	1.9088	10.9386	6.3808	3.9500	1.0000	0.0322
77.6	0.0000	0.0000	1.3876	15.0473	8.7776	5.4338	1.0000	0.0443
91.9	0.0000	0.0000	1.1728	17.8033	10.3852	6.4290	1.0000	0.0524
124.9	0.0000	0.0000	0.8626	24.2067	14.1206	8.7413	1.0000	0.0713
145.0	0.0000	0.0000	0.7428	28.1081	16.3964	10.1501	1.0000	0.0828
174.8	0.0000	0.0000	0.6163	33.8778	19.7620	12.2336	1.0000	0.0997
221.1	0.0000	0.0000	0.4873	42.8511	24.9965	15.4740	1.0000	0.1262
271.0	0.0000	0.0000	0.3975	52.5222	30.6380	18.9664	1.0000	0.1546
333.7	0.0706	0.0080	0.3229	64.6643	37.7209	23.3510	0.9294	0.1904
427.2	0.2079	0.0234	0.2522	82.7952	48.2972	29.8983	0.7921	0.2438
503.1	0.2765	0.0312	0.2141	97.5111	56.8815	35.2123	0.7235	0.2871
623.2	0.3667	0.0413	0.1729	120.7739	70.4515	43.6128	0.6333	0.3556
709.9	0.4216	0.0475	0.1518	137.5927	80.2624	49.6862	0.5784	0.4051
804.8	0.4667	0.0526	0.1339	155.9755	90.9857	56.3245	0.5333	0.4592
904.9	0.4922	0.0555	0.1191	175.3815	102.3059	63.3322	0.5078	0.5164
1004.0	0.5158	0.0582	0.1073	194.5822	113.5063	70.2658	0.4842	0.5729
1119.6	0.5373	0.0606	0.0962	216.9923	126.5788	78.3583	0.4627	0.6389
1230.5	0.5589	0.0630	0.0876	238.4818	139.1144	86.1184	0.4411	0.7021
1325.1	0.5746	0.0648	0.0813	256.8065	149.8038	92.7357	0.4254	0.7561



Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction <i>V_p</i>	J Function
	fraction <i>V_p</i>	fraction <i>V_b</i>		A/B [Lab]	O/B [Lab]	O/B [Res]		
1421.7	0.5883	0.0663	0.0758	275.5303	160.7260	99.4971	0.4117	0.8112
1517.8	0.6020	0.0679	0.0710	294.1670	171.5974	106.2270	0.3980	0.8661
1611.8	0.6138	0.0692	0.0668	312.3734	182.2178	112.8015	0.3862	0.9197
1713.7	0.6256	0.0705	0.0629	332.1283	193.7415	119.9352	0.3744	0.9779
1912.1	0.6471	0.0730	0.0563	370.5762	216.1694	133.8192	0.3529	1.0911
2079.4	0.6628	0.0747	0.0518	403.0043	235.0858	145.5293	0.3372	1.1865

Pore Diameter Distribution

Range Pore Diameter	Incremental % <i>pv</i>	Frequency fraction
200 - 30	0.2218	0.0842
30 - 10	0.0277	0.0105
10 - 5	0.0000	0.0000
5 - 1	0.0069	0.0026
1 - 0.5	0.0347	0.0132
0.5 - 0.1	1.9751	0.7500
0.1 - 0.05	0.3673	0.1395

TABLE 5A: WELL DUYONG DEEP-1
SAMPLE NO: 2 DEPTH: 3184.0 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION [0-2,000 psia]



Sample ID **3**
 Depth **3528.0 meter**
 Permeability to air **0.005 md**
 Porosity **0.093 fraction Vp**

Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction <i>Vp</i>	J Function
	fraction <i>Vp</i>	fraction <i>Vb</i>		A/B [Lab]	O/B [Lab]	O/B [Res]		
0.5	0.0000	0.0000	211.2451	0.0988	0.0577	0.0357	1.0000	0.0001
1.0	0.0000	0.0000	109.9337	0.1899	0.1108	0.0686	1.0000	0.0001
1.5	0.0000	0.0000	72.7939	0.2868	0.1673	0.1036	1.0000	0.0002
2.1	0.0000	0.0000	51.3024	0.4070	0.2374	0.1470	1.0000	0.0003
3.1	0.0000	0.0000	34.9789	0.5969	0.3482	0.2156	1.0000	0.0004
4.1	0.0000	0.0000	26.5357	0.7869	0.4590	0.2841	1.0000	0.0006
5.6	0.0000	0.0000	19.3768	1.0776	0.6286	0.3891	1.0000	0.0008
7.0	0.0000	0.0000	15.3250	1.3625	0.7948	0.4920	1.0000	0.0010
8.5	0.0000	0.0000	12.6301	1.6532	0.9644	0.5970	1.0000	0.0012
10.5	0.0000	0.0000	10.2507	2.0369	1.1882	0.7356	1.0000	0.0014
13.0	0.0000	0.0000	8.3001	2.5156	1.4675	0.9084	1.0000	0.0018
16.0	0.0000	0.0000	6.7292	3.1029	1.8100	1.1205	1.0000	0.0022
20.0	0.0000	0.0000	5.4003	3.8665	2.2555	1.3962	1.0000	0.0027
23.0	0.0000	0.0000	4.6882	4.4537	2.5980	1.6083	1.0000	0.0031
25.0	0.0000	0.0000	4.3129	4.8413	2.8241	1.7483	1.0000	0.0034
29.9	0.0000	0.0000	3.6008	5.7988	3.3826	2.0940	1.0000	0.0041
41.2	0.0000	0.0000	2.6175	7.9772	4.6534	2.8806	1.0000	0.0056
67.2	0.0000	0.0000	1.6027	13.0278	7.5996	4.7045	1.0000	0.0091
70.3	0.0000	0.0000	1.5327	13.6228	7.9466	4.9194	1.0000	0.0095
80.8	0.0000	0.0000	1.3335	15.6578	9.1337	5.6542	1.0000	0.0109
87.6	0.0000	0.0000	1.2293	16.9854	9.9082	6.1336	1.0000	0.0119
109.6	0.0000	0.0000	0.9833	21.2337	12.3863	7.6677	1.0000	0.0148
123.8	0.0000	0.0000	0.8704	23.9877	13.9928	8.6622	1.0000	0.0168
144.7	0.0000	0.0000	0.7445	28.0461	16.3602	10.1278	1.0000	0.0196
200.8	0.0000	0.0000	0.5366	38.9149	22.7003	14.0526	1.0000	0.0272
222.9	0.0000	0.0000	0.4833	43.2019	25.2011	15.6007	1.0000	0.0302
278.6	0.0000	0.0000	0.3867	53.9913	31.4949	19.4969	1.0000	0.0378
337.5	0.0000	0.0000	0.3193	65.4027	38.1516	23.6177	1.0000	0.0457
426.7	0.0000	0.0000	0.2525	82.6924	48.2373	29.8612	1.0000	0.0578
528.6	0.0000	0.0000	0.2038	102.4532	59.7644	36.9970	1.0000	0.0716
620.0	0.0000	0.0000	0.1738	120.1615	70.0942	43.3917	1.0000	0.0840
718.4	0.0000	0.0000	0.1500	139.2226	81.2132	50.2748	1.0000	0.0974
802.4	0.0000	0.0000	0.1343	155.5045	90.7110	56.1544	1.0000	0.1087
904.8	0.0000	0.0000	0.1191	175.3660	102.2969	63.3266	1.0000	0.1226
1003.8	0.0000	0.0000	0.1073	194.5473	113.4859	70.2532	1.0000	0.1361
1115.1	0.0000	0.0000	0.0966	216.1182	126.0690	78.0427	1.0000	0.1511
1231.2	0.0124	0.0011	0.0875	238.6078	139.1879	86.1639	0.9876	0.1669



Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction <i>V_p</i>	J Function
	fraction <i>V_p</i>	fraction <i>V_b</i>		A/B [Lab]	O/B [Lab]	O/B [Res]		
1323.5	0.0371	0.0034	0.0814	256.5119	149.6319	92.6293	0.9629	0.1794
1419.8	0.0544	0.0050	0.0759	275.1737	160.5180	99.3683	0.9456	0.1924
1518.0	0.0840	0.0078	0.0710	294.1941	171.6132	106.2368	0.9160	0.2057
1609.6	0.1137	0.0105	0.0669	311.9606	181.9770	112.6524	0.8863	0.2182
1711.9	0.1433	0.0133	0.0629	331.7872	193.5426	119.8121	0.8567	0.2320
1921.7	0.1828	0.0169	0.0561	372.4328	217.2525	134.4896	0.8172	0.2605
2071.7	0.2199	0.0203	0.0520	401.5061	234.2119	144.9883	0.7801	0.2808

Pore Diameter Distribution

Range Pore Diameter	Incremental <i>% pv</i>	Frequency <i>fraction</i>
200 - 30	0.0680	0.0916
30 - 10	0.0227	0.0305
10 - 5	0.0000	0.0000
5 - 1	0.0057	0.0076
1 - 0.5	0.0057	0.0076
0.5 - 0.1	0.0737	0.0992
0.1 - 0.05	0.5670	0.7634

TABLE 5B: WELL DUYONG DEEP-1
SAMPLE NO: 3 DEPTH: 3528.0 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION [0-2,000 psia]



Sample ID 4
Depth 3576.0 meter
Permeability to air 0.038 md
Porosity 0.112 fraction Vp

Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction <i>Vp</i>	J Function
	fraction <i>Vp</i>	fraction <i>Vb</i>		A/B [Lab]	O/B [Lab]	O/B [Res]		
0.5	0.0000	0.0000	211.2451	0.0988	0.0577	0.0357	1.0000	0.0002
1.0	0.0000	0.0000	109.9337	0.1899	0.1108	0.0686	1.0000	0.0003
1.5	0.0000	0.0000	72.7939	0.2868	0.1673	0.1036	1.0000	0.0005
2.1	0.0000	0.0000	51.3024	0.4070	0.2374	0.1470	1.0000	0.0007
3.1	0.0000	0.0000	34.9789	0.5969	0.3482	0.2156	1.0000	0.0010
4.1	0.0000	0.0000	26.5357	0.7869	0.4590	0.2841	1.0000	0.0014
5.6	0.0000	0.0000	19.3768	1.0776	0.6286	0.3891	1.0000	0.0019
7.0	0.0000	0.0000	15.3250	1.3625	0.7948	0.4920	1.0000	0.0024
8.5	0.0000	0.0000	12.6301	1.6532	0.9644	0.5970	1.0000	0.0029
10.5	0.0000	0.0000	10.2507	2.0369	1.1882	0.7356	1.0000	0.0036
13.0	0.0000	0.0000	8.3001	2.5156	1.4675	0.9084	1.0000	0.0044
16.0	0.0000	0.0000	6.7334	3.1009	1.8089	1.1198	1.0000	0.0054
20.0	0.0000	0.0000	5.4003	3.8665	2.2555	1.3962	1.0000	0.0068
23.0	0.0000	0.0000	4.6882	4.4537	2.5980	1.6083	1.0000	0.0078
25.0	0.0000	0.0000	4.3129	4.8413	2.8241	1.7483	1.0000	0.0085
29.9	0.0000	0.0000	3.6008	5.7988	3.3826	2.0940	1.0000	0.0101
41.3	0.0000	0.0000	2.6111	7.9966	4.6647	2.8876	1.0000	0.0140
55.2	0.0000	0.0000	1.9521	10.6963	6.2395	3.8626	1.0000	0.0187
62.3	0.0000	0.0000	1.7290	12.0762	7.0445	4.3609	1.0000	0.0211
72.9	0.0000	0.0000	1.4787	14.1209	8.2372	5.0992	1.0000	0.0247
88.4	0.0000	0.0000	1.2190	17.1288	9.9918	6.1854	1.0000	0.0300
93.2	0.0000	0.0000	1.1556	18.0688	10.5401	6.5248	1.0000	0.0316
117.1	0.0000	0.0000	0.9201	22.6931	13.2376	8.1947	1.0000	0.0397
155.5	0.0000	0.0000	0.6928	30.1392	17.5812	10.8836	1.0000	0.0527
178.8	0.0000	0.0000	0.6024	34.6608	20.2188	12.5164	1.0000	0.0607
223.1	0.0000	0.0000	0.4830	43.2329	25.2192	15.6119	1.0000	0.0757
268.9	0.0000	0.0000	0.4007	52.1055	30.3949	18.8159	1.0000	0.0912
325.9	0.0000	0.0000	0.3305	63.1681	36.8481	22.8107	1.0000	0.1105
410.7	0.0000	0.0000	0.2623	79.5915	46.4284	28.7414	1.0000	0.1393
518.5	0.0000	0.0000	0.2078	100.4822	58.6146	36.2852	1.0000	0.1758
628.8	0.0773	0.0087	0.1713	121.8631	71.0868	44.0061	0.9227	0.2133
717.4	0.1587	0.0178	0.1502	139.0443	81.1092	50.2104	0.8413	0.2433
797.5	0.2162	0.0243	0.1351	154.5665	90.1638	55.8157	0.7838	0.2705
903.4	0.2677	0.0301	0.1192	175.0947	102.1386	63.2286	0.7323	0.3064
1004.4	0.3054	0.0343	0.1073	194.6597	113.5515	70.2938	0.6946	0.3406
1118.8	0.3411	0.0383	0.0963	216.8411	126.4907	78.3037	0.6589	0.3795
1225.7	0.3649	0.0410	0.0879	237.5496	138.5706	85.7818	0.6351	0.4157



Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction <i>V_p</i>	J Function
	fraction <i>V_p</i>	fraction <i>V_b</i>		A/B [Lab]	O/B [Lab]	O/B [Res]		
1316.9	0.3768	0.0423	0.0818	255.2269	148.8824	92.1653	0.6232	0.4466
1413.4	0.3867	0.0434	0.0762	273.9275	159.7911	98.9183	0.6133	0.4794
1519.3	0.3966	0.0445	0.0709	294.4461	171.7602	106.3278	0.6034	0.5153
1615.4	0.4065	0.0457	0.0667	313.0692	182.6237	113.0528	0.5935	0.5479
1710.1	0.4165	0.0468	0.0630	331.4384	193.3391	119.6861	0.5835	0.5800
1910.1	0.4343	0.0488	0.0564	370.2002	215.9501	133.6834	0.5657	0.6478
2074.3	0.4482	0.0503	0.0519	402.0236	234.5138	145.1752	0.5518	0.7035

Pore Diameter Distribution

Range Pore Diameter	Incremental % <i>pv</i>	Frequency fraction
200 - 30	0.3280	0.1326
30 - 10	0.0428	0.0173
10 - 5	0.0357	0.0144
5 - 1	0.1070	0.0432
1 - 0.5	0.0428	0.0173
0.5 - 0.1	1.4046	0.5677
0.1 - 0.05	0.5134	0.2075

**TABLE 5C: WELL DUYONG DEEP-1
SAMPLE NO: 4 DEPTH: 3576.0 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION [0-2,000 psia]**



Sample ID 6
Depth 3598.5 meter
Permeability to air 0.010 md
Porosity 0.100 fraction V_p

Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction V_p	J Function
	fraction V_p	fraction V_b		A/B [Lab]	O/B [Lab]	O/B [Res]		
0.5	0.0000	0.0000	211.2451	0.0988	0.0577	0.0357	1.0000	0.0001
1.0	0.0000	0.0000	109.9337	0.1899	0.1108	0.0686	1.0000	0.0002
1.5	0.0000	0.0000	72.7939	0.2868	0.1673	0.1036	1.0000	0.0003
2.1	0.0000	0.0000	51.3024	0.4070	0.2374	0.1470	1.0000	0.0004
3.1	0.0000	0.0000	34.9789	0.5969	0.3482	0.2156	1.0000	0.0006
4.1	0.0000	0.0000	26.5357	0.7869	0.4590	0.2841	1.0000	0.0007
5.6	0.0000	0.0000	19.3768	1.0776	0.6286	0.3891	1.0000	0.0010
7.0	0.0000	0.0000	15.3033	1.3644	0.7959	0.4927	1.0000	0.0013
8.5	0.0000	0.0000	12.6301	1.6532	0.9644	0.5970	1.0000	0.0016
10.5	0.0000	0.0000	10.2410	2.0389	1.1893	0.7363	1.0000	0.0019
13.0	0.0000	0.0000	8.3001	2.5156	1.4675	0.9084	1.0000	0.0024
16.0	0.0000	0.0000	6.7334	3.1009	1.8089	1.1198	1.0000	0.0030
20.0	0.0000	0.0000	5.4003	3.8665	2.2555	1.3962	1.0000	0.0037
23.0	0.0000	0.0000	4.6882	4.4537	2.5980	1.6083	1.0000	0.0042
25.0	0.0000	0.0000	4.3129	4.8413	2.8241	1.7483	1.0000	0.0046
29.9	0.0000	0.0000	3.6008	5.7988	3.3826	2.0940	1.0000	0.0055
38.1	0.0000	0.0000	2.8307	7.3764	4.3029	2.6637	1.0000	0.0070
46.7	0.0000	0.0000	2.3075	9.0489	5.2785	3.2677	1.0000	0.0086
63.5	0.0000	0.0000	1.6966	12.3069	7.1790	4.4441	1.0000	0.0117
74.3	0.0000	0.0000	1.4496	14.4039	8.4023	5.2014	1.0000	0.0137
77.3	0.0000	0.0000	1.3934	14.9853	8.7414	5.4114	1.0000	0.0143
87.1	0.0000	0.0000	1.2376	16.8711	9.8415	6.0923	1.0000	0.0161
95.3	0.0000	0.0000	1.1311	18.4603	10.7685	6.6662	1.0000	0.0176
116.6	0.0000	0.0000	0.9241	22.5962	13.1811	8.1597	1.0000	0.0215
146.9	0.0000	0.0000	0.7332	28.4783	16.6123	10.2838	1.0000	0.0271
178.9	0.0000	0.0000	0.6023	34.6647	20.2210	12.5178	1.0000	0.0330
221.8	0.0000	0.0000	0.4858	42.9791	25.0711	15.5202	1.0000	0.0409
272.9	0.0000	0.0000	0.3949	52.8808	30.8471	19.0958	1.0000	0.0504
331.7	0.0000	0.0000	0.3248	64.2825	37.4981	23.2131	1.0000	0.0612
424.4	0.0000	0.0000	0.2539	82.2428	47.9750	29.6988	1.0000	0.0783
523.0	0.0000	0.0000	0.2060	101.3524	59.1222	36.5995	1.0000	0.0965
624.5	0.0000	0.0000	0.1725	121.0395	70.6064	43.7087	1.0000	0.1153
701.1	0.0000	0.0000	0.1537	135.8872	79.2675	49.0704	1.0000	0.1294
798.4	0.0000	0.0000	0.1349	154.7390	90.2644	55.8780	1.0000	0.1474
902.5	0.0753	0.0075	0.1194	174.9164	102.0346	63.1643	0.9247	0.1666
1003.1	0.2030	0.0203	0.1074	194.4000	113.4000	70.2000	0.7970	0.1851
1117.9	0.3033	0.0303	0.0964	216.6493	126.3787	78.2345	0.6967	0.2063



Injection Pressure <i>psia</i>	Mercury Saturation		Pore Throat Radius <i>microns</i>	Equivalent Injection Pressure <i>psia</i>			Wetting Phase Saturation fraction <i>V_p</i>	J Function
	fraction <i>V_p</i>	fraction <i>V_b</i>		A/B [Lab]	O/B [Lab]	O/B [Res]		
1229.2	0.3763	0.0375	0.0876	238.2260	138.9652	86.0261	0.6237	0.2269
1315.8	0.4265	0.0426	0.0819	255.0060	148.7535	92.0855	0.5735	0.2429
1413.2	0.4538	0.0453	0.0762	273.8849	159.7662	98.9029	0.5462	0.2608
1519.5	0.4880	0.0487	0.0709	294.5003	171.7919	106.3473	0.5120	0.2805
1608.2	0.5086	0.0507	0.0670	311.6796	181.8131	112.5510	0.4914	0.2968
1706.7	0.5177	0.0517	0.0631	330.7678	192.9479	119.4439	0.4823	0.3150
1913.9	0.5359	0.0535	0.0563	370.9289	216.3752	133.9465	0.4641	0.3533
2074.3	0.5496	0.0548	0.0519	402.0216	234.5126	145.1745	0.4504	0.3829

Pore Diameter Distribution

Range Pore Diameter	Incremental % <i>pv</i>	Frequency fraction
200 - 30	0.3218	0.1614
30 - 10	0.0316	0.0158
10 - 5	0.0000	0.0000
5 - 1	0.0379	0.0190
1 - 0.5	0.0000	0.0000
0.5 - 0.1	0.6436	0.3228
0.1 - 0.05	0.9591	0.4810

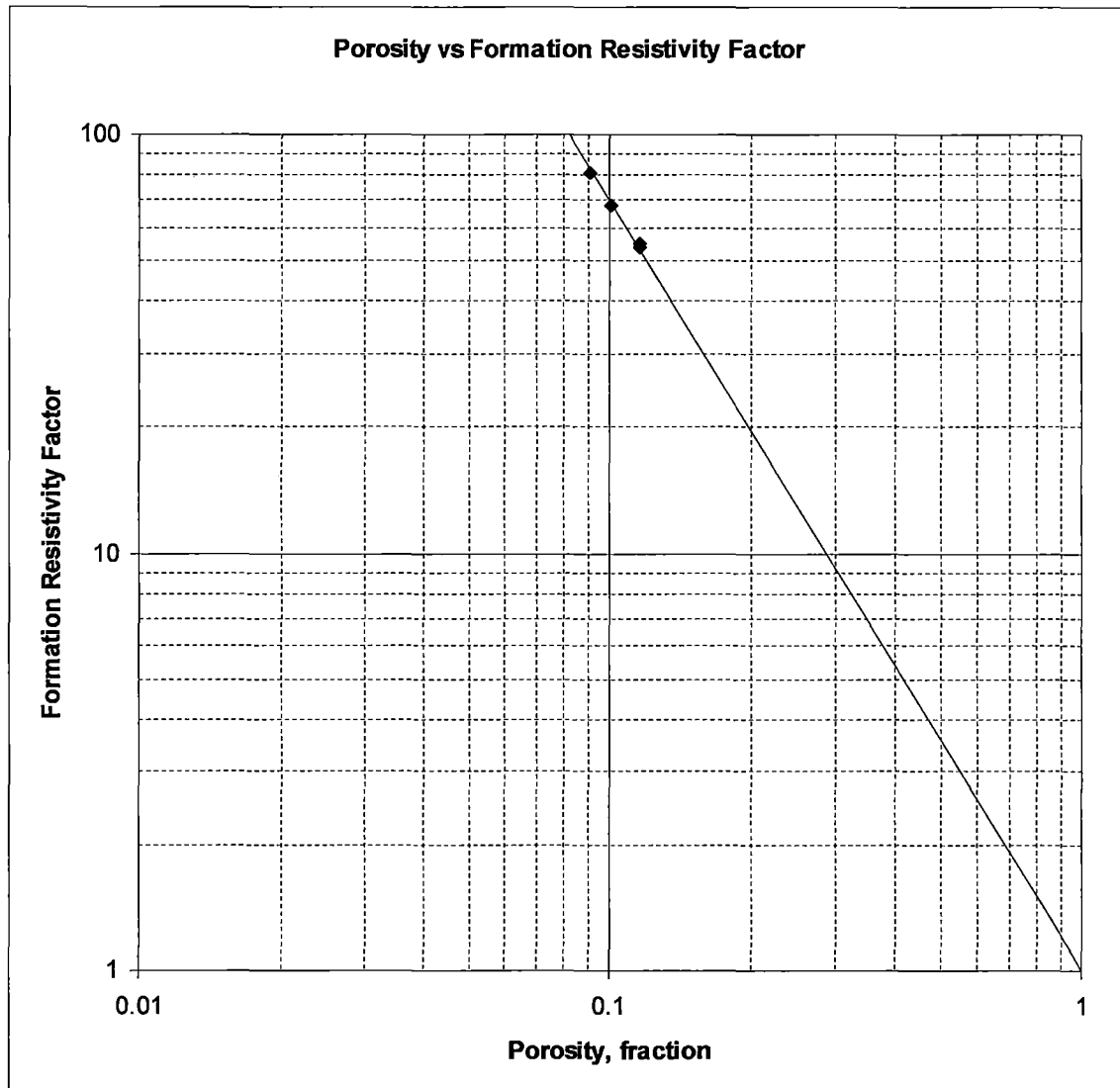
TABLE 5D: WELL DUYONG DEEP-1
SAMPLE NO: 6 DEPTH: 3598.5 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION [0-2,000 psia]



GRAPHS



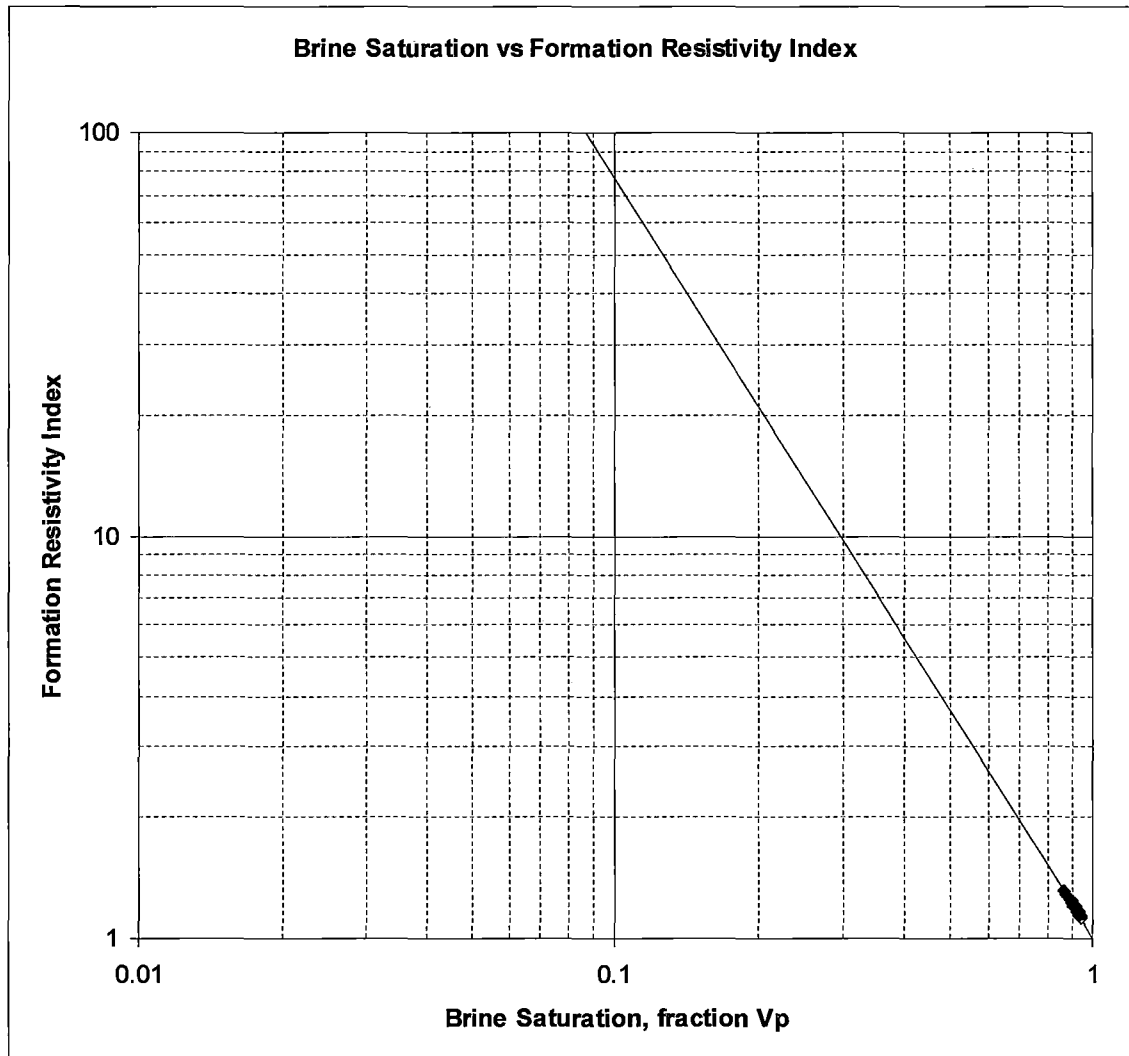
Saturant	3,500 ppm
Resistivity of saturant	1.500 ohm-m @ 77° F
Porosity Exponent [m] Composite	1.84
Y-Intercept [a] Composite	1.00
Net Overburden Pressure	560 psi



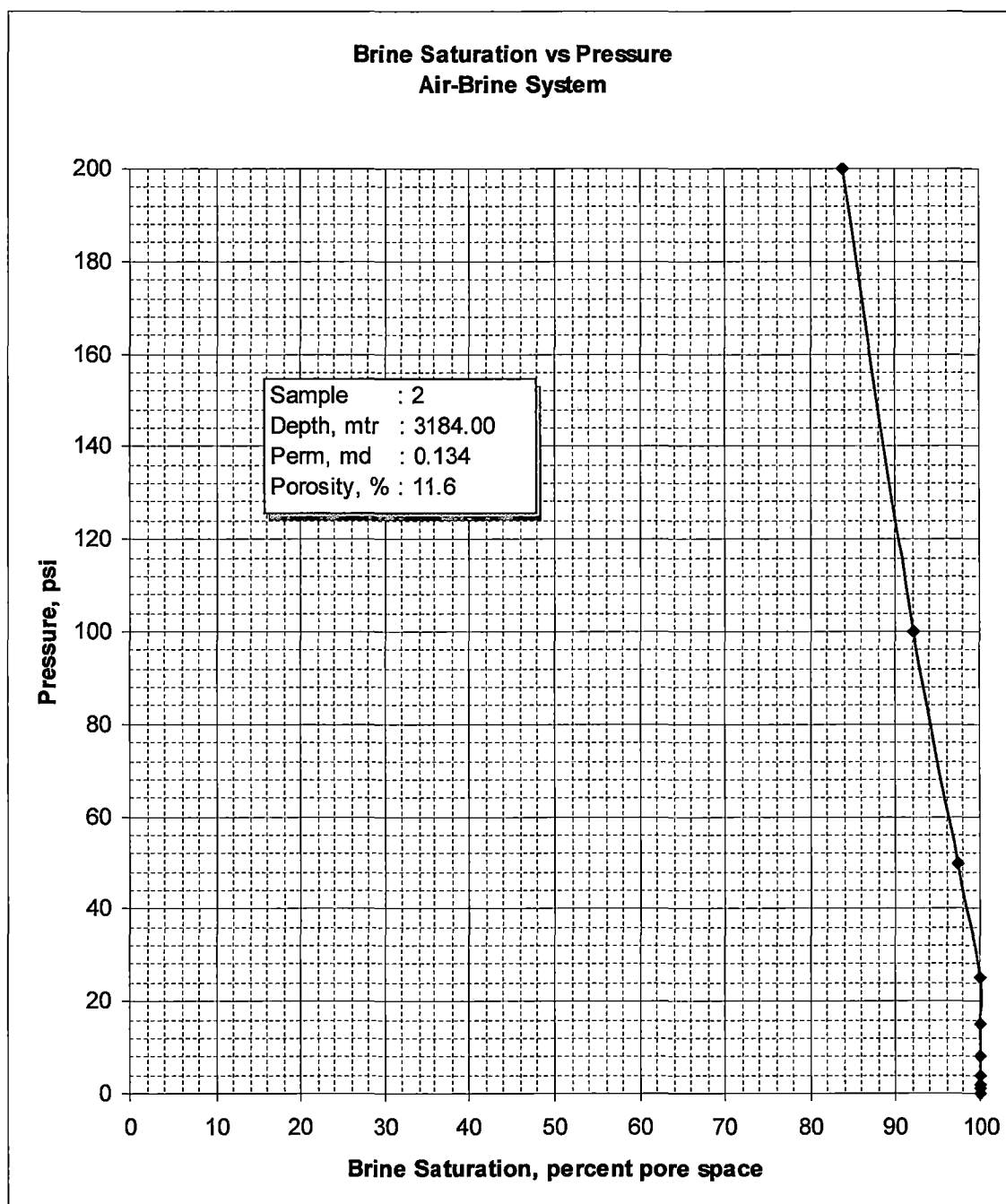
**GRAPH 1: WELL DUYONG DEEP-1
FORMATION RESISTIVITY FACTOR AT NOB CONDITIONS**



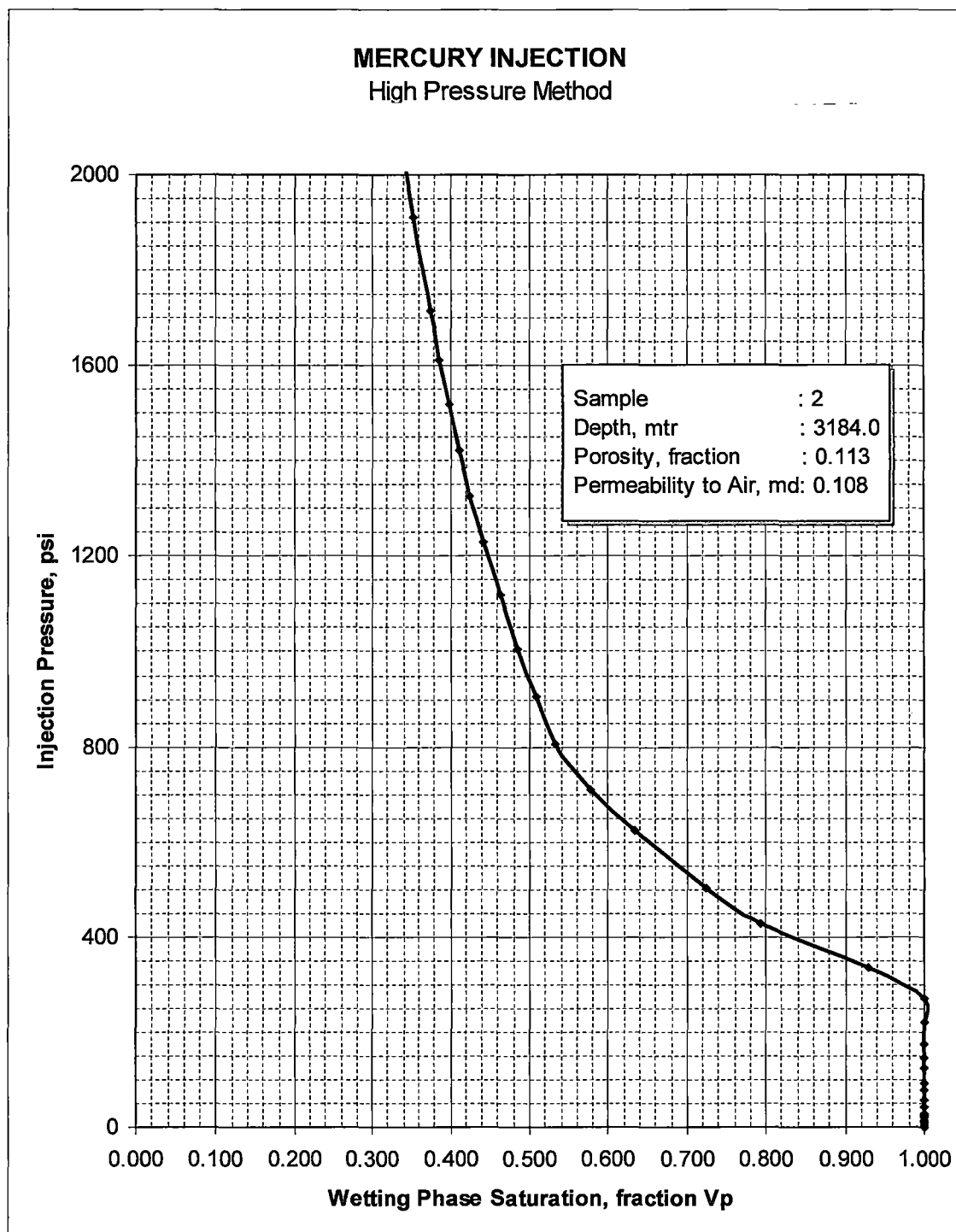
Saturant	3,500 ppm
Resistivity of saturant	1.500 ohm-m @ 77° F
Porosity Exponent [m] Composite	1.84
Y-Intercept [a] Composite	1.00
Saturant Exponent [n] Composite	1.88
Net Overburden Pressure	560 psi



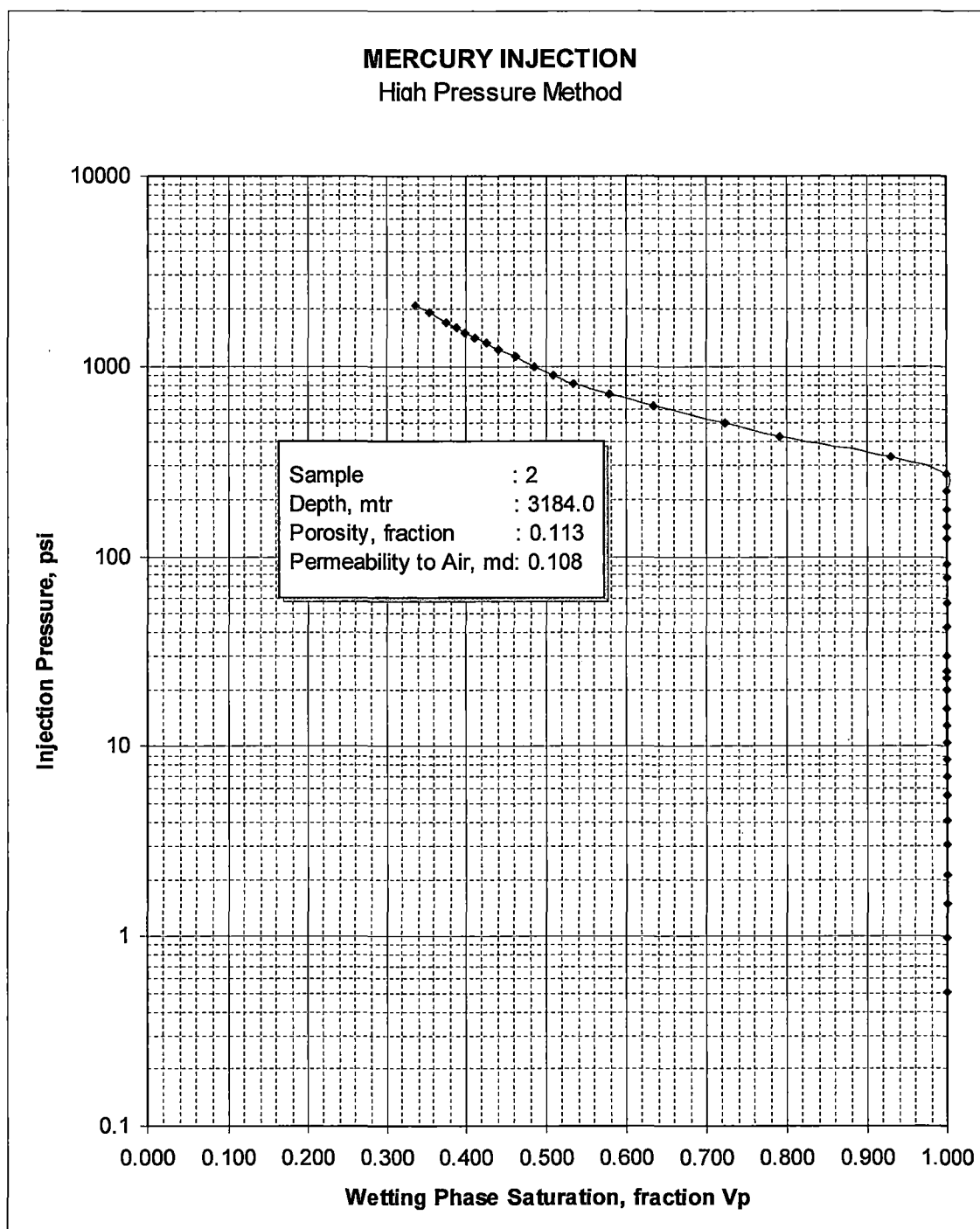
**GRAPH 2: WELL DUYONG DEEP-1
FORMATION RESISTIVITY INDEX AT NOB CONDITIONS**



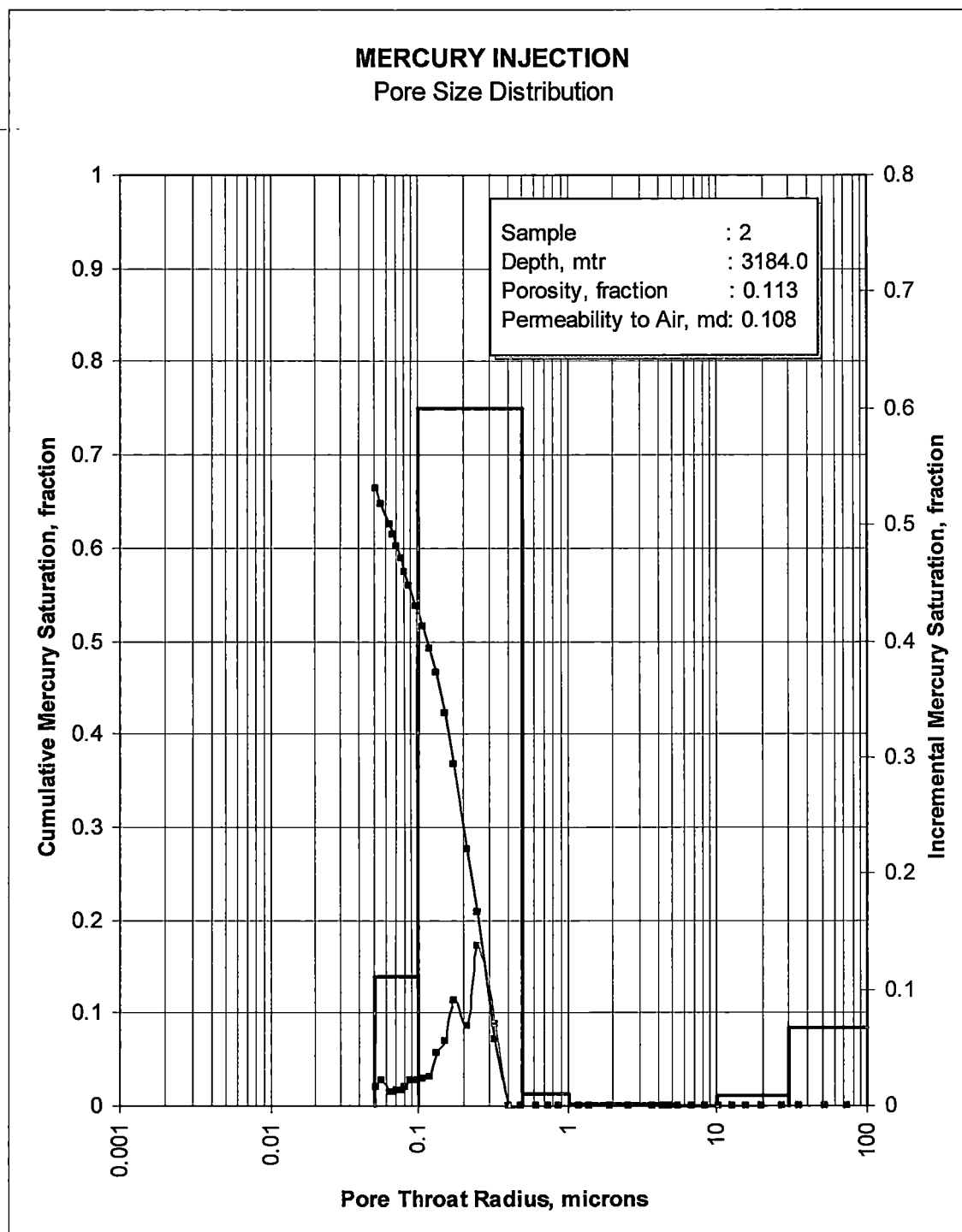
**GRAPH 3: WELL DUYONG DEEP-1
SAMPLE NO: 2 DEPTH: 3184.0 M
AIR-BRINE CAPILLARY PRESSURE BY CENTRIFUGE**



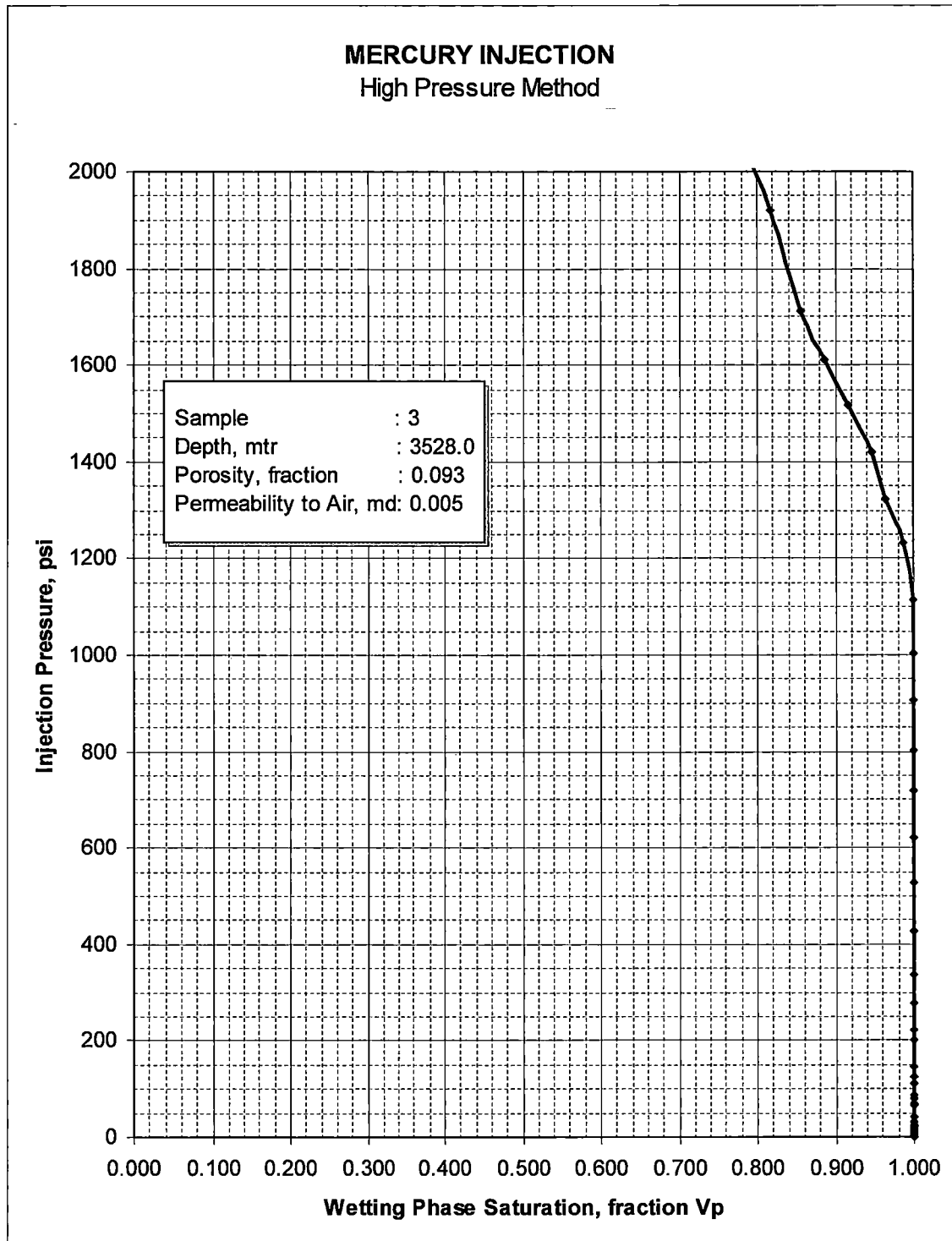
GRAPH 4A: WELL DUYONG DEEP-1
SAMPLE NO: 2 DEPTH: 3184.0 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [LINEAR]



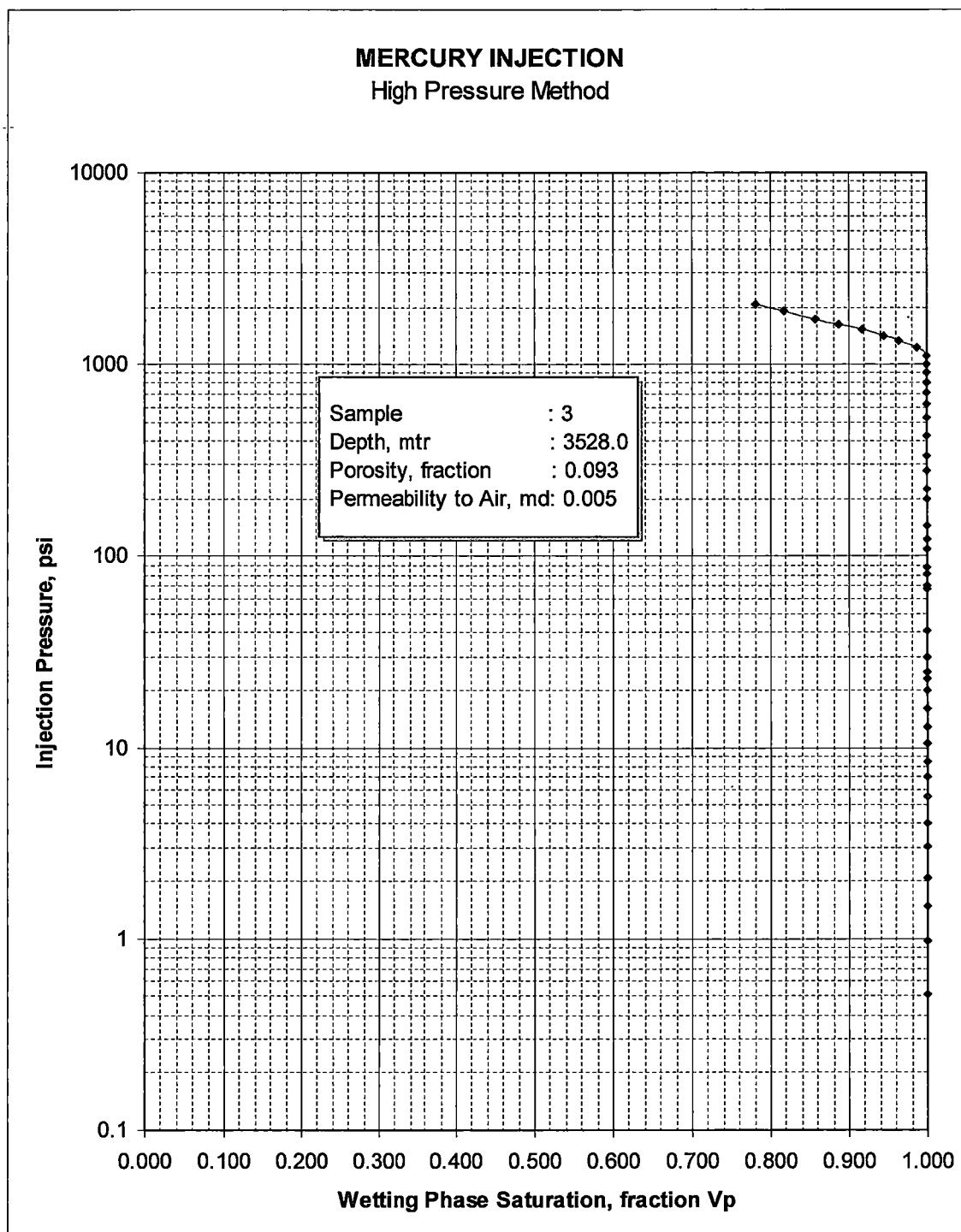
GRAPH 5A: WELL DUYONG DEEP-1
SAMPLE NO: 2 DEPTH: 3184.0 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [SEMILOG]



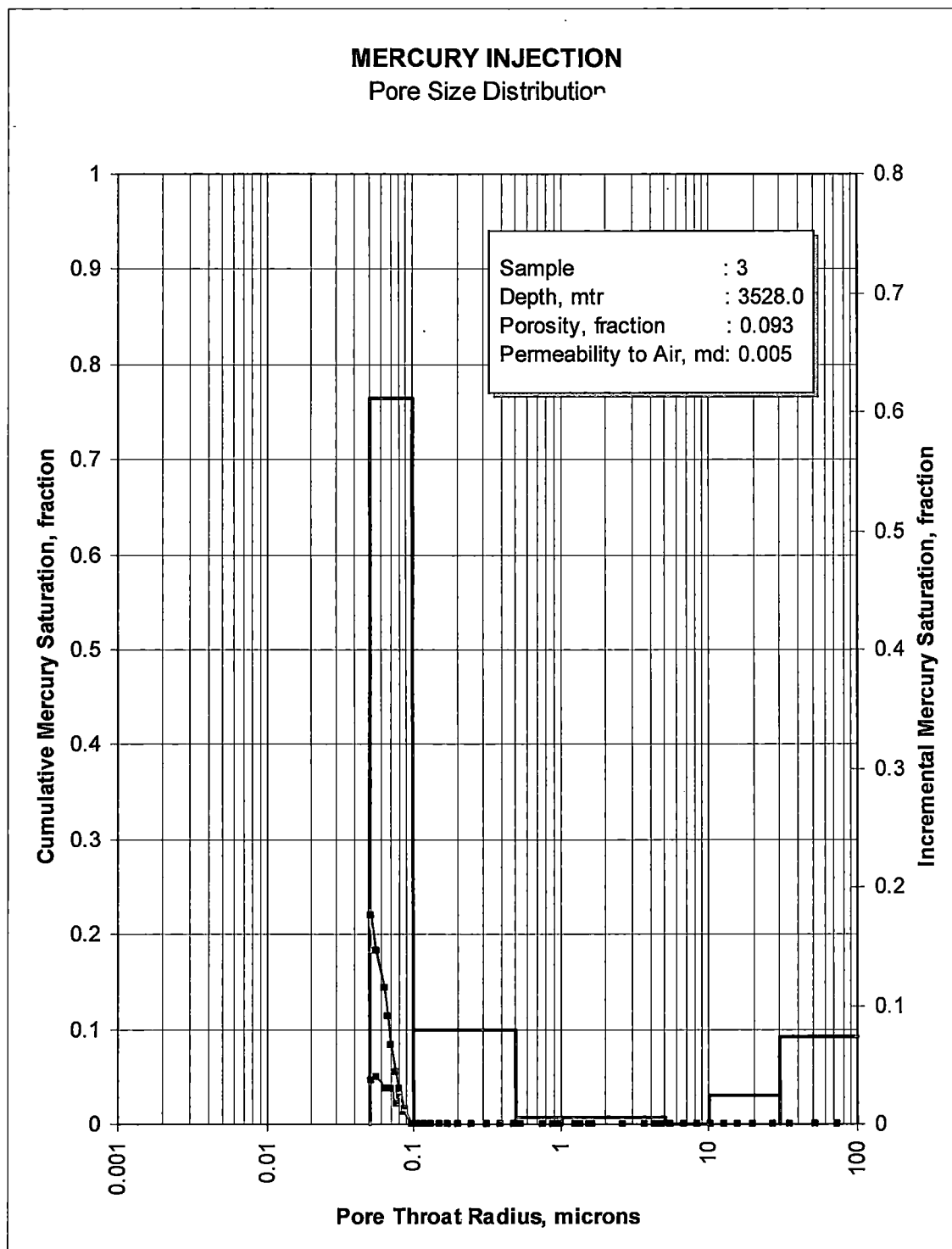
GRAPH 6A: WELL DUYONG DEEP-1
SAMPLE NO: 2 DEPTH: 3184.0 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION HISTOGRAM



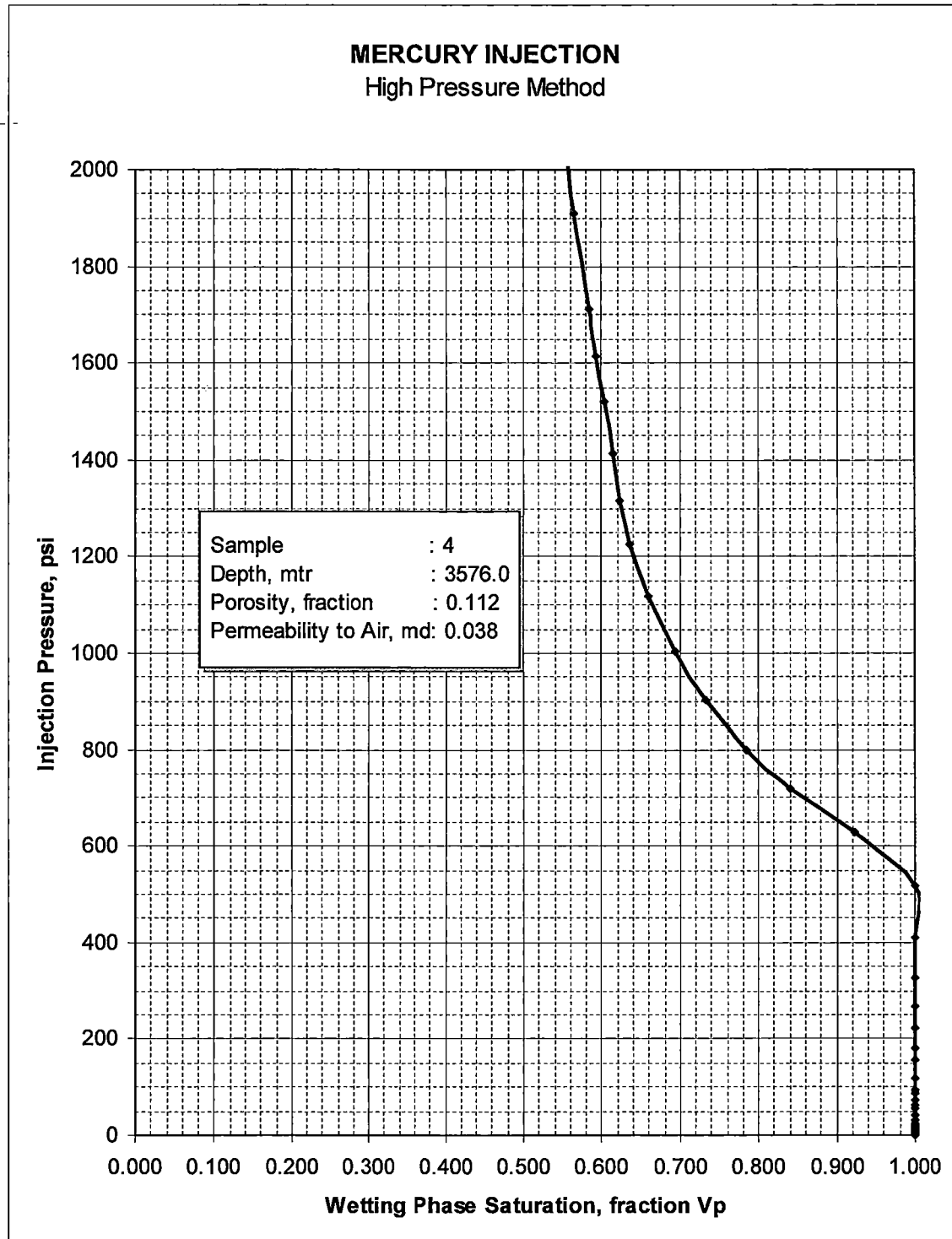
GRAPH 4B: WELL DUYONG DEEP-1
SAMPLE NO: 3 DEPTH: 3528.0 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [LINEAR]



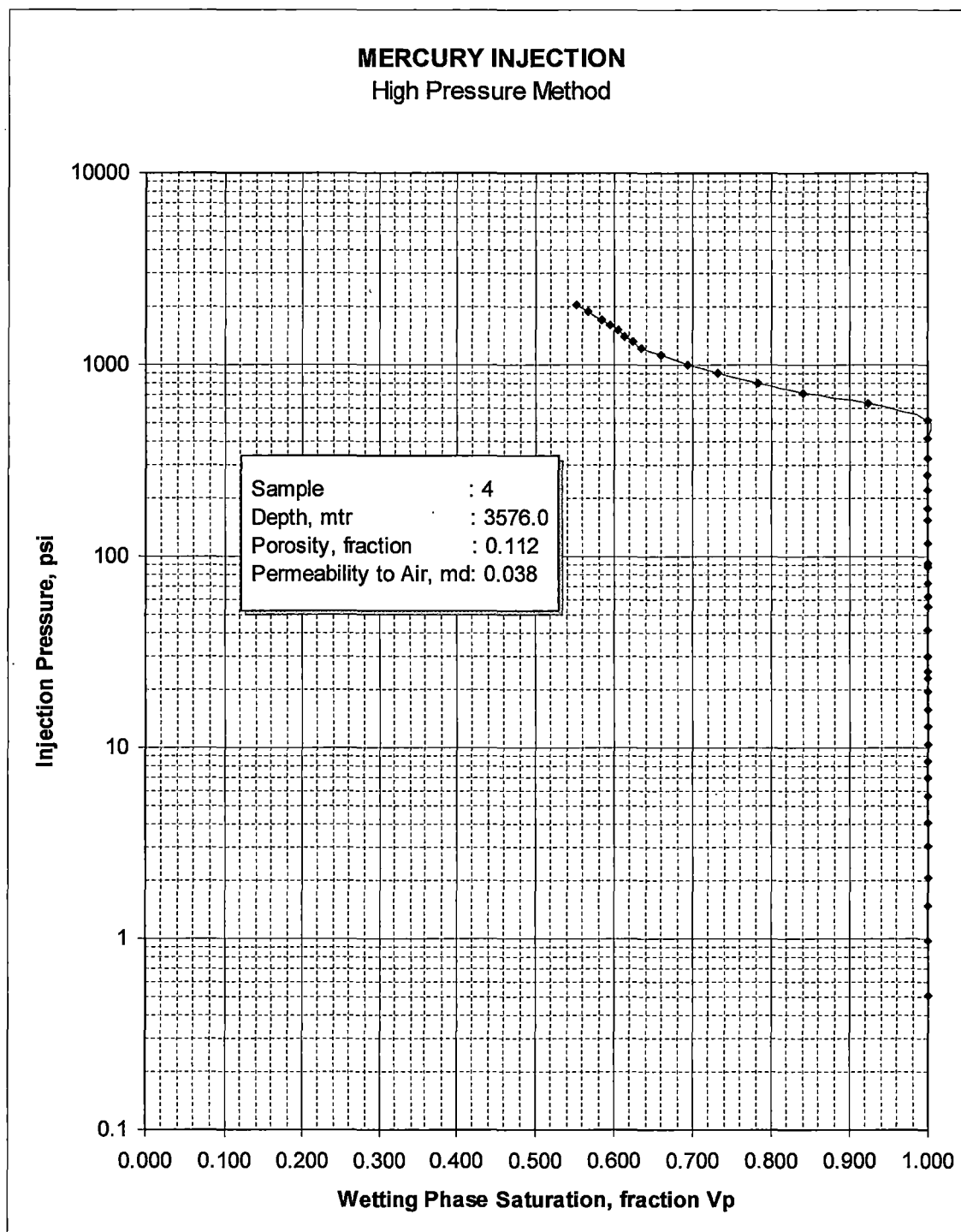
GRAPH 5B: **WELL DUYONG DEEP-1**
SAMPLE NO: 3 **DEPTH: 3528.0 M**
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [SEMILOG]



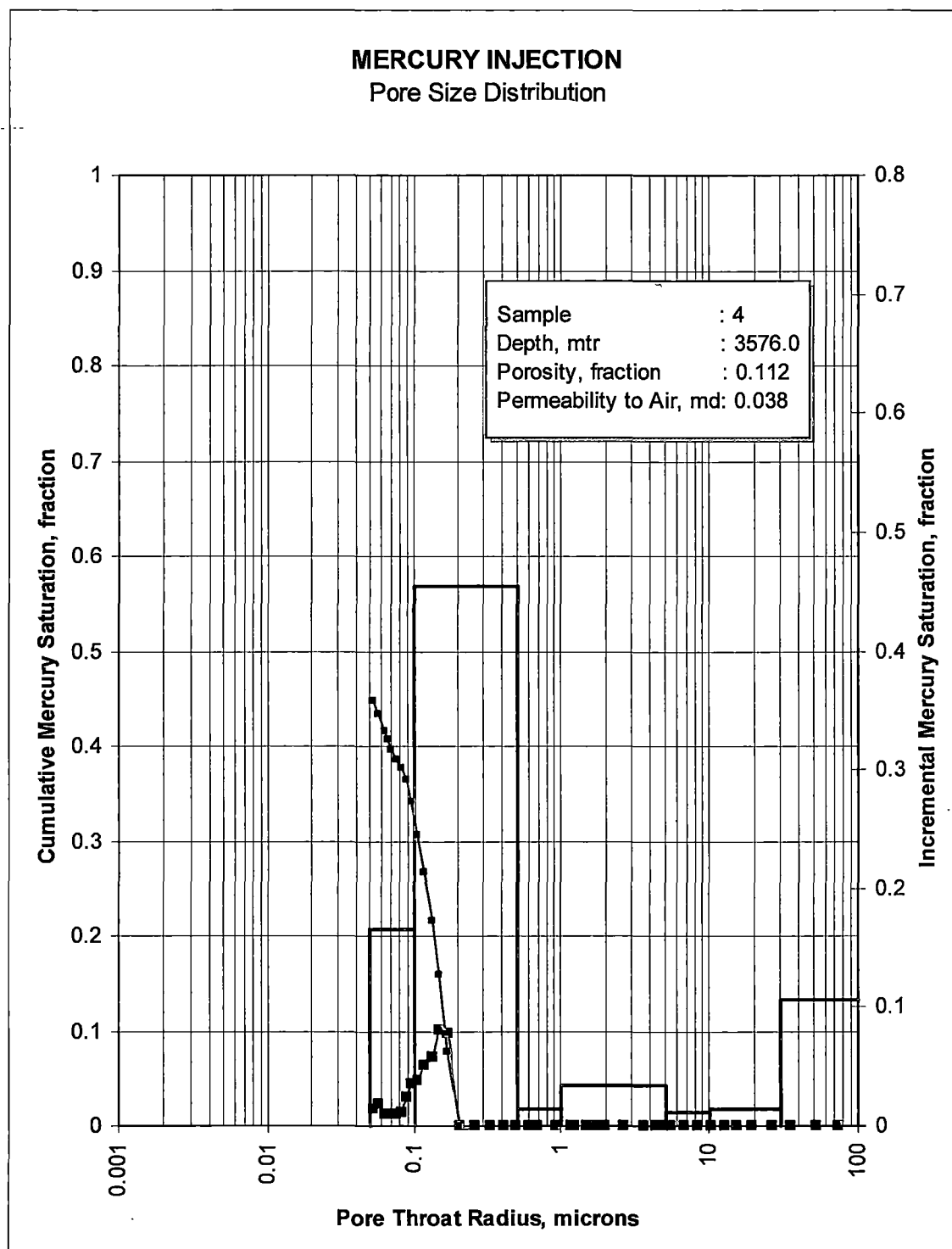
GRAPH 6B: WELL DUYONG DEEP-1
SAMPLE NO: 3 DEPTH: 3528.0 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION HISTOGRAM



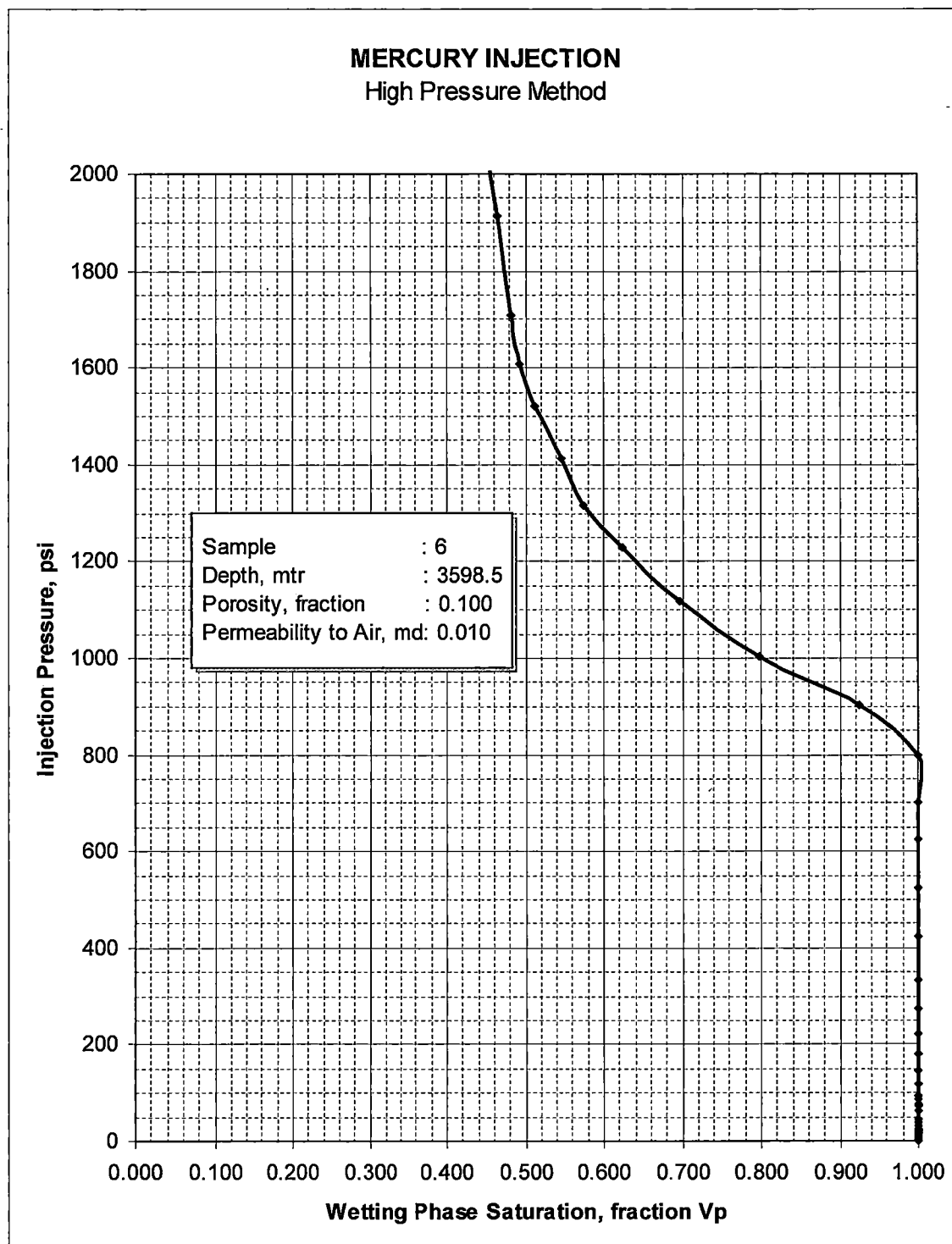
GRAPH 4C: WELL DUYONG DEEP-1
SAMPLE NO: 4 DEPTH: 3576.0 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [LINEAR]



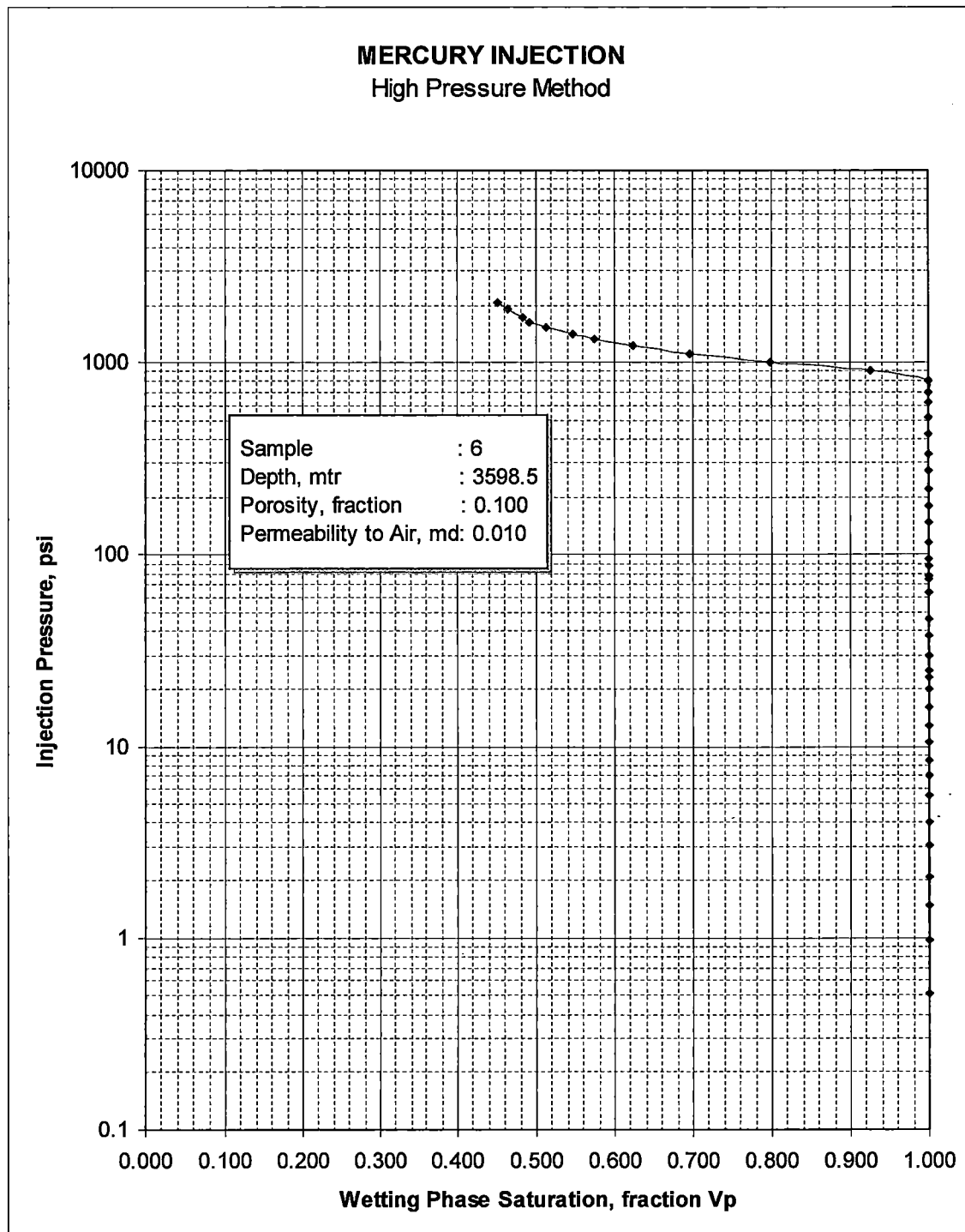
GRAPH 5C: **WELL DUYONG DEEP-1**
SAMPLE NO: 4 DEPTH: 3576.0 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [SEMILOG]



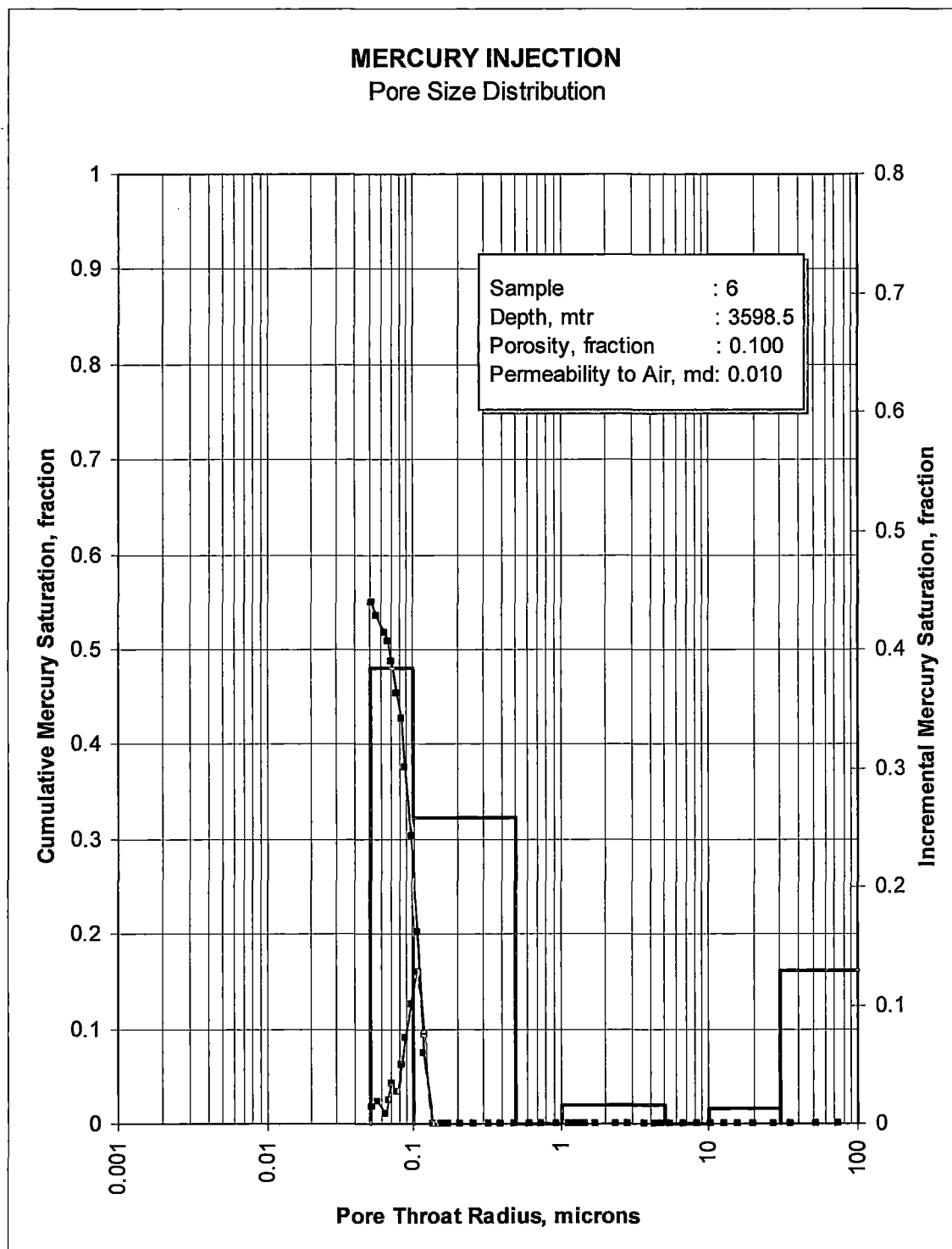
GRAPH 6C: WELL DUYONG DEEP-1
SAMPLE NO: 4 DEPTH: 3576.0 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION HISTOGRAM



GRAPH 4D: **WELL DUYONG DEEP-1**
SAMPLE NO: 6 DEPTH: 3598.5 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [LINEAR]



GRAPH 5D: WELL DUYONG DEEP-1
SAMPLE NO: 6 DEPTH: 3598.5 M
MERCURY INJECTION CAPILLARY PRESSURE VS. WETTING
PHASE SATURATION [SEMILOG]



GRAPH 6D: WELL DUYONG DEEP-1
SAMPLE NO: 6 DEPTH: 3598.5 M
MERCURY INJECTION CAPILLARY PRESSURE & PORE SIZE
DISTRIBUTION HISTOGRAM